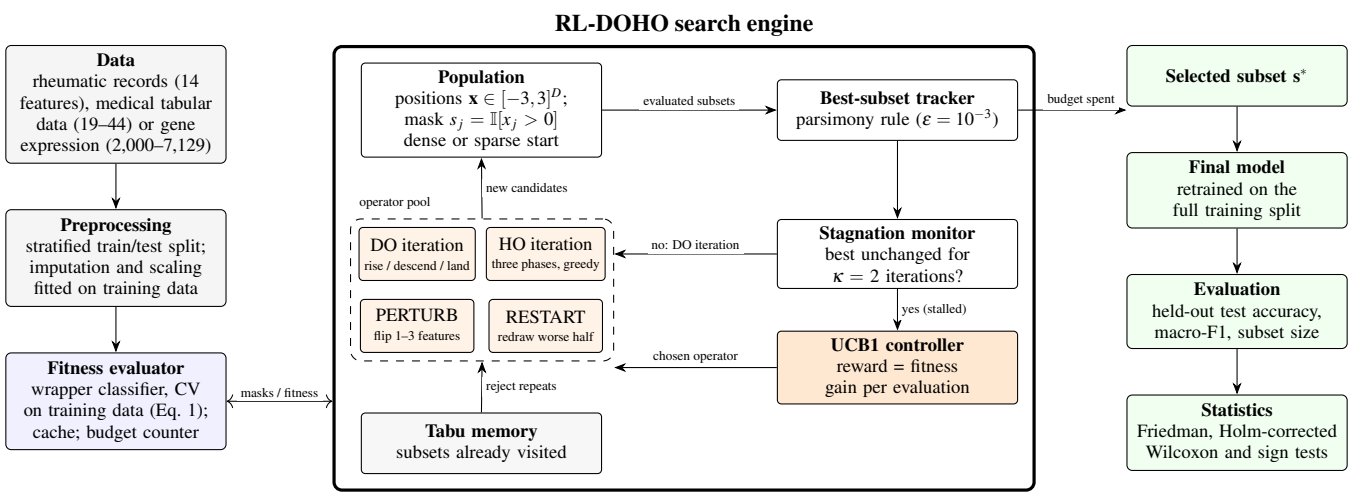

RL-DOHO search engine
Data
rheumatic records (14 features), medical tabular data (19–44) or gene expression (2,000–7,129)
Preprocessing
stratified train/test split; imputation and scaling fitted on training data
Fitness evaluator
wrapper classifier, CV on training data (Eq. 1); cache; budget counter
masks / fitness
Population
positions $\mathbf{x} \in [-3,3]^D$; mask $s_j = \mathbb{I}[x_j > 0]$ dense or sparse start
new candidates
operator pool
DO iteration
rise / descend / land
HO iteration
three phases, greedy
PERTURB
flip 1–3 features
RESTART
redraw worse half
reject repeats
Tabu memory
subsets already visited
evaluated subsets
Best-subset tracker
parsimony rule ($\varepsilon = 10^{-3}$)
Stagnation monitor
best unchanged for $\kappa = 2$ iterations?
no: DO iteration
yes (stalled)
UCB1 controller
reward = fitness gain per evaluation
chosen operator
budget spent
Selected subset $\mathbf{s}^*$
Final model
retrained on the full training split
Evaluation
held-out test accuracy, macro-F1, subset size
Statistics
Friedman, Holm-corrected Wilcoxon and sign tests